CUHK-Shenzhen
Course Outline

1. Course Identity

A. Course as listed in CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	MAT2002
Course title (English)	Ordinary Differential Equations
Course title (Chinese)	常微分方程
Units	3
Description (English)	This course teaches fundamental principles and methods of ordinary differential equations with some important theory and sophisticated techniques. It covers the topics of first order differential equations, high order differential equations, systems of linear equations, Laplace transforms, and some basic theory of nonlinear differential equations, as well as their applications.
Description (Chinese)	本课程教授常微分方程的基本原理和方法，包括一些重要的理论和精巧的技术。课程内容包括一阶方程，高阶方程，线性方程组，Laplace 变换，非线性方程的基本理论，以及它们的应用。

B. Corresponding course in CUHK

Please give details of the *closest* corresponding course in CUHK (as approved by CUHK Senate and listed in course list). If the course in SZ maps to more than one course in CUHK, please make multiple copies of the block below.

Course code	MATH3270
Course title (English)	Ordinary Differential Equations
Course title (Chinese)	常微分方程
Units	3
Description (English)	Ordinary differential equations arise from quantitative description of natural and social phenomena. Topics are: method of explicit solution, linear equations and systems, series solutions, Sturm-Liouville boundary value problems, dynamical systems and stability, applications to mechanics, electrical networks and population of species.
Description (Chinese)	

1. Prerequisites

(MAT1001 or MAT1010 or MAT1011) & (MAT1002B or MAT 1002 or MAT1012 or MAT3010) & (MAT2040 or MAT2041 or MAT2042 or MAT2041A)

A. Co-requisites

None.

2. Learning Outcomes

Upon successful completion of the courses, the students will be able to understand basic theory of ODE (ordinary differential equations), master basic techniques in ODE, and handle nonlinear ODE by using qualitative theory of ODE.

Knowledge

- The basic concept, geometric understanding of ordinary differential equations.
- Elementary Integration method about separable equation, first order nonlinear equation, exact equation and implicit equation.
- The basic theory of ordinary differential equations, existence and uniqueness of solutions of ordinary differential equations.
- The general theory of systems of ordinary differential equations
- The stability theory of nonlinear ordinary differential equations and its applications
- The properties of basic boundary value problems.

Skill

- Ability of carrying out mathematical modelling, solving some simple first order ordinary differential equations.
- Ability of solving higher order linear equations.
- Ability of solving systems of ordinary differential equations.
- Ability of analyzing stability of nonlinear ordinary differential equations, solving basic boundary value problems.

3. Course syllabus

Introduction, some basic concept about ordinary differential equations and solutions of differential equations, classification of differential equations, geometric explanation of ordinary differential equations.
Elementary integration method: linear first order ordinary differential equations, separable equations, exact equations, exact equations with integrating factors, implicit equations.
Existence and Uniqueness Theorem for first-order initial value problems.
Systems of linear ordinary differential equations: basic theory, Wronskian determinant, Liouville formula, the method of variations of constants, high order linear ordinary differential equations, undetermined coefficient method, solving systems using linear algebra, eigenvalues.
Laplace transforms and their applications to solve initial value problems.
Stability: phase portraits for linear systems, concept of stability, method of linearization.
Stability: Liapunov's function, applications to population models, Liapunov's second method (if time permits).
Stability: Periodic solutions, limit cycles, Poincare-Bendixon Theorem, Lotka-Volterra Competition Systems, Prey-Predator System (if time permits).

4. Assessment Scheme

Component/ method	% weight
Assignments	20%
Mid-Term Exam	30%
Final Exam	50%

5. Descriptor

Grade	Overall course
A	Demonstrates the ability to synthesize and apply the principles or subject matter learnt in the course, to novel situations and/or in novel ways, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.
A-	Demonstrates the ability to state and partially apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is usually logically persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.
B	Demonstrates the ability to state and apply the principles or subject matter learnt in the course to most (but not all) familiar and standard situations in a manner that is not incorrect but is somewhat fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.
C	Demonstrates the ability to state and sometimes apply the principles or subject matter learnt in the course to some simple and familiar situations in a manner that is broadly correct in its essentials. Has the ability to state the knowledge or application in simple terms.
D	Unsatisfactory performance on a number of learning outcomes, OR failure to meet specified assessment requirements.
F	Demonstrates the ability to synthesize and apply the principles or subject matter learnt in the course, to novel situations and/or in novel ways, in a manner that would surpass the normal expectation at this level, and typical of standards that may be common at higher levels of study. Has the ability to express the synthesis of ideas or application in a clear and cogent manner.

6. Declaration on the use of AI tools.

Prohibit the use of AI tools: all AI tools are not allowed in the assessments. Any student who violates this rule will be considered to be engaging in academic dishonesty. Students are required to complete all assessments independently without any external assistance.

7. Feedback for evaluation

A course questionnaire (CTE) was answered by each student after completing all teaching activities.
Reflections from teachers of more advanced courses.
Feedback from informal conversations over in the instructor's office and wechat/e-mail correspondence with students after class.

8. Reading

A. Required

Elementary differential equations and boundary value problems. William E. Boyce and Richard C. DiPrima, 10th edition. Wiley.

B. Recommended

Ordinary differential equation, Bin Liu, Peking University press, (Chinese series)
Ordinary differential equation, Weinian Zhang, Zhengdong Du & Bin Xu, Higher Education press, (Chinese series)
Ordinary differential equation, Tongren Ding & Chengzhi Li, Higher Education press, (Chinese series)
Basic Theory of Ordinary Differential Equations, Hsieh, Po-Fang, Sibuya, Yasutaka, 1999, 1st edition, Springer-Verlag New York.
Differential Equations and their Applications, by Martin Braun, 4th edition, Springer, 1993.
Differential Equations and linear algebra, C. Henry Edwards, David E. Penney and David T. Calvis, 4th edition. Pearson.

9. Course components

Activity	**Hours/week**
Lecture	3 hours per week
Tutorial	1 hour per week

10. Indicative teaching plan

Week	**Content/ topic/ activity**
1	**Definition of differential equations, classification of differential equations, mathematical models by using ODE (Population model and pendulum model)**
2	**First order linear differential equations, method of integrating factors, separable equations, exact and non-exact equations**
3	**Existence and uniqueness theorem for first-order ODE, Picard iterative method**
4	**Second-order differential equations: mechanical vibration model, existence and uniqueness theorem for linear equation, Wronskian, fundamental set of solutions**
5	**Second-order linear differential equations with constant coefficients: eigenvalue method for homogeneous equation, variation of parameters for non-homogeneous equation, Euler's equation.**

6	**Second-order linear differential equations with constant coefficients: Euler's equation. High-order linear differential equations.**
7	**System of first order differential equations: SIR model, Lotka-Volterra model, existence and uniqueness theorem for linear system of equations, Wronskian, fundamental set of solutions.**
8	**System of first order differential equations: fundamental matrix, method of variation of parameters.**
9	**Laplace transform and its application.**
10	**Geometrical approach for a single ODE, the phase plane, phase portrait for 2 by 2 linear system.**
11	**Stability of equilibrium solution, qualitative properties of orbits, stability and asymptotic stability.**
12	**Stability and asymptotic stability: long time behavior of solutions. Linear approximation for nonlinear system, application to competing species and pendulum system.**
13	**Liapunov's second method, application to the damped pendulum system (if time permits).**
14	**Periodic solutions, limit cycle, Poincare-Bendixon Theorem, application to Predator-Prey model (if time allows) .**

11. Implementation plan (2025-2026)

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections.

Expected enrolment: 240
Number of lecturing sections: 2
Number of tutorial sections: 5

12. Approval

Has the course title been included in the program submission approved by CUHK Senate? Are there any differences?

Yes. The course syllabus might be slightly different.

Have the details (as in this document) been approved at School or other level in CUHK (SZ)?

13. Any other information

14. Version date

Version number	MAT2002 v004 251010
As of (date)	Oct. 10, 2025